



Over 15+ published award winning papers

Dr. Shuyi Sun

Senior HCI Researcher

Ph.D.

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Please visit my website



MISSION STATEMENT

I'm Shuyi, a results-driven PhD in Computer Science from UC Davis, where I built tech that makes the invisible visible—biosensing systems that pick up the subtle signals our bodies (and even our pets) are constantly giving off. I pioneered Animal Biosensing Computing to help monitor pet health non-invasively, and I've turned everyday items—like jewelry and cosmetics—into wearable diagnostics for people. At the end of the day, I'm a builder: equal parts scientific rigor and creative spark, driven to make ideas real.



EDUCATION

University of California, Davis

2020 - 2025

Ph.D. - Computer Science

Virginia Polytechnic Institute

2019 - 2020

Ph.D. - Computer Science

Georgia Institute of Technology

2015 - 2019

B.Sc. - Computer Science



LANGUAGE



English

Native Speaker



Chinese

Native Speaker



Japanese

Business Level



Korean

Conversational



SKILLS

Electronics & Embedded Prototyping

ESP32 | Arduino | CircuitPython | Custom PCB Design | Electrochemical & Colorimetric Biosensors | Wearable IoT | 3D Printing | Laser Cutting | Silicone Molding | Resin Casting | E-Textiles

Programming & Systems

Python (ML/DL, TensorFlow, PyTorch) | C / C++ for Embedded Systems | JS | Web UI | Firebase | Supabase | API Design | Sensor-to-Cloud Pipelines | Interactive Visualization | Cross-platform

Creativity & Productivity

HCI | Machine Learning | Deep Learning | Computer Vision for Biosensing | Rapid UX/UI Prototyping | Fusion 360 | Adobe Illustrator | LaTeX | Digital Fabrication Workflows | Mixed-Materials Design



RECENT WORK

Ph.D. Researcher

2020 - 2025

Interactive Organisms Lab, UC Davis

- Led the fabrication of cutting-edge biosensors & microfluidics
- Designed custom ESP-32 PCBs with software in C & MicroPython
- Published award winning papers on HCI and Machine Learning

Graduate Researcher

2019-2020

TriM Lab, Virginia Polytechnic Institute

- Conducted independent hardware and software research on VR
- Managed lab projects & logistics, mentoring junior researchers.
- Worked with 3D game engines such as Unity & Unreal Engine

Undergraduate Researcher

2017-2019

FIDO Lab, Georgia Institute of Technology

- Developed an experimental device for measuring dog temperament
- Fabricated embedded systems for smart wearable devices
- Designed interactive prototypes using CAD and silicone casting

Instructor

2019 - XXX

UC Davis, GA Tech

- Intro to Programming; Web Programming @ UC Davis
- Prototyping Intelligent Appliances @ GA Tech



PUBLICATIONS

DiabetiCAT: Animal Biosensing Computing w/Feline Urine Glucose Biosensing & Hydration Monitoring

cirCAT: Cat Centered Smart Home System and Veterinary Complementary Devices

pawH: Colorimetric pH-Sensing Toys for Non-Invasive Pet Health Monitoring

PURRentio: Implementing a Smart Litter Box for Feline Urinalysis with Electrochemical Biosensors

Nail pHolish: Integrating Chemical Detection into Cosmetic Fingernail Products

ChromaLipSense: Designing a Lipstick-Based Biosensors for Metabolic Monitoring

BioSparks: Jewelry as Electrochemical Sweat Biosensors with Modular, Repurposing and Interchangeable Approaches

BioCosMe: Lip-based Cosmetics with Colorimetric Biosensors for Salivary Analysis using Deep Learning

WOOFlex: A Wearable Device to Aid Canine Flexibility Exercises

GemiN' I: Seamless Skin Interfaces Aiding Communication through Unconscious Behaviors

GluCAT: A Feline Biofluids IoT Hub for Electrochemical Glucose Biosensing

Technology for Working Dogs

See CV for full history..

SHUYI SUN, Ph.D.

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Education

2020–2025 **University of California, Davis**, *Ph.D.*, Davis, CA.
Computer Science Engineering

2015–2019 **Georgia Institute of Technology**, *BS*, Atlanta, GA.
Computer Science [Media and People]

Technical Skills

Electronics Prototyping.

- 3D Modeling
- Sewing & Embroidery
- Silicone Molding
- 3D Printing
- Laser Cutting
- Resin Crafts

Programming.

- Python
- Web UI Development
- C, C++, C#
- Java
- HTML, CSS, JavaScript
- Arduino, CircuitPython

Creativity and Productivity.

- Deep learning
- Google Firebase
- Autodesk Software
- Machine Learning
- Embrilliance
- LaTeX

Languages

English
(Native and
Trilingual
Proficiency)



Chinese
(Native and
Trilingual
Proficiency)



Japanese
(Trilingual
Proficiency)



Korean
(Elementary
proficiency)



Arts Creativity

Over 15 years training and practicing on Procreate & Digital Art, Charcoal Drawing, and Acrylic Painting.

Work Experiences

- 2020-2025 **Graduate Researcher**, *Interactive Organisms Lab (IO Lab)*, University of California, Davis.
Conducting research in: Wearability factors for body-Worn colorimetric biosensors; Biosensors for metabolic monitoring; Animal biosensing computing; Designing microfluidic fingernails and integrating chemical detection, etc. Prototyping and integrating AI models into ubiquitous computing. Assisting the lab in hosting symposiums and academic events.
- 2023 **Instructor**, *Intro to Programming*, University of California, Davis.
- 2022 **Instructor**, *Web Programming*, University of California, Davis.
- 2021 **Lead Teaching Assistant**, *Coding for Designers*, University of California, Davis.
- 2019-2020 **Research Assistant**, *TriM Lab*, Virginia Polytechnic Institute.
Conduct independent research in addition to aiding advisors in lab projects, lab management, and logistics. Assist the lab in hosting symposiums and academic events.
- 2018-2019 **Teaching Assistant**, *Prototyping Intelligent Appliances*, Georgia Institute of Technology.
Held office hours to help students with workshops and projects, graded and logged student work, wrote guidelines for future TAs of the course, and assisted the professor in course planning.

Publications & Conference Presentations

- 2025 **DiabetiCAT: Animal Biosensing Computing with Feline Urine Glucose Biosensing and Hydration Monitoring**.
Shuyi Sun, Jinho Yon, Xingda Chen, Krystle Reagan, and Katia Vega. (ACI '25) In Proceedings of the Twelfth International Conference on Animal-Computer Interaction.
- 2025 **pawH: Colorimetric pH-Sensing Toys for Non-Invasive Pet Health Monitoring**, <https://doi.org/10.1145/3715336.3735768>.
Shuyi Sun, Yuan-Hao Ku, and Katia Vega. (DIS '25) In Designing Interactive Systems Conference (DIS '25), July 05–09, 2025, Funchal, Portugal. Association for Computing Machinery, New York, NY, USA 15 Pages.
- 2025 **Nail pHolish: Integrating Chemical Detection into Cosmetic Fingernail Products**.
Shuyi Sun, Dana Mayfield, Jinho Yon, Yuan-Hao Ku, and Katia Vega. (INTERACT '25) In Proceedings of the 2025 ACM 20th IFIP TC13 International Conference on Human-Computer Interaction
- 2025 **ChromaLipSense: Designing a Lipstick-Based Biosensors for Metabolic Monitoring**, <https://doi.org/10.1080/14606925.2025.2516592>.
Sun, S., Ku, Y. H., Suk, J. Y., Yetisen, A. K., Vega, K. (2025). The Design Journal. 1-23.
- 2024 **Wearability Factors for Body-Worn Colorimetric Biosensors, Best Demo Award on Conference Presentation**, <https://doi.org/10.1145/3675094.3678486>.
Sun, S., Yetisen, A., & Vega, K. (2024, October). In Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing (pp. 944-951).

- 2024 **BioCosMe: Lip-based Cosmetics with Colorimetric Biosensors for Salivary Analysis using Deep Learning**, <https://doi.org/10.1145/3675095.3676610>.
Sun, S., Ku, Y. H., Unsihuay, N., Florez, O., Suk, J. Y., Yetisen, A. K., & Vega, K. (2024, October). In Proceedings of the 2024 ACM International Symposium on Wearable Computers (pp. 32-39).
- 2024 **ChromaLipSense: Lipstick-Based Biosensors for Metabolic Monitoring**.
Sun, S., Kua, Y. H., Suk, J. Y., Yetisen, A. K., Vega, K. (2024). In WEARABLES COLLECTIVE (WC '24), March 25-26, 2024.
- 2024 **GluCAT: A Feline Biofluids IoT Hub for Electrochemical Glucose Biosensing**, <https://doi.org/10.1145/3623509.3635250>.
Sun, S., Reagan, K., Seker, E., & Vega, K. (2024, February). In Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction (pp. 1-7).
- 2023 **cirCAT: Cat Centered Smart Home System and Veterinary Complementary Devices**, <https://doi.org/10.1145/3637882.3637901>.
Sun, S., Vega, K. (2023, December). In Proceedings of the Tenth International Conference on Animal-Computer Interaction. Article 21, 1-6.
- 2023 **cirCAT: PURRtentio: a Litter Box that Monitors Feline Urine using Electrochemical Biosensors**, <https://doi.org/10.1145/3637882.3637887>.
Sun, S., Vega, G., Reagan, K., Seker, E., & Vega, K. (2023, December). In Proceedings of the Tenth International Conference on Animal-Computer Interaction (pp. 1-13).
- 2023 **PURRtentio: Implementing a Smart Litter Box for Feline Urinalysis with Electrochemical Biosensors**, <https://doi.org/10.1145/3586182.3615820>.
Sun, S., Vega, G., Seker, E., Reagan, K., & Vega, K. (2023, October). In Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (pp. 1-3).
- 2023 **BioSparks: Jewelry as Electrochemical Sweat Biosensors with Modular, Repurposing and Interchangeable Approaches**, <https://doi.org/10.1145/3594739.3610787>.
Sun, S., Ruiz, A., Pirmoradi, S., & Vega, K. (2023, October). In Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (pp. 315-320).
- 2021 **WOOFlex: A Wearable Device to Aid Canine Flexibility Exercises**, <https://doi.org/10.1145/3493842.3493903>.
Sun, S., Vega, G., Marcellin-Little, D., & Vega, K. (2021, November). In Proceedings of the Eight International Conference on Animal-Computer Interaction (pp. 1-4).
- 2021 **GemiN' I: Seamless Skin Interfaces Aiding Communication through Unconscious Behaviors, Best Demo Award on Conference Presentation**, <https://doi.org/10.1145/3410531.3414298>.
Sun, S., Deshmukh, N., Chen, X., Wang, H. C., & Vega, K. (2021, February). In Proceedings of the Augmented Humans International Conference 2021 (pp. 277-279).
- 2018 **Technology for Working Dogs**, <https://doi.org/10.1145/3295598.3295615>.
Jackson, M. M., Byrne, C., Freil, L., Valentin, G., Zuerndorfer, J., Zeagler, C., ... & Starner, T. (2018, December). In Proceedings of the Fifth International Conference on Animal-Computer Interaction (pp. 1-5).

Research Projects in Process

- 2024 **Lab-on-a-nail: Designing Microfluidic Fingernails through a Women-Centered Approach**, Shuyi Sun, Dana Mayfield, Yuan-Hao Ku, and Katia Vega.

Honors & Awards

- 2024 **Best Demo Award**, *Wearability Factors for Body-Worn Colorimetric Biosensors*, ACM International Joint Conference on Pervasive and Ubiquitous Computing.
- 2021 **Best Demo Award**, *GemiN' I: Seamless Skin Interfaces Aiding Communication through Unconscious Behaviors*, The Augmented Humans (AHs) International Conference.
- 2019 **Graduate with Honor**, Georgia Institute of Technology, May 2019.

Professional Activities

- 2021 **Chair, Gadget Show**, ACM International Symposium on Wearable Computers, September 21–26, 2021.
- 2019 **VR Viewfinder: The Effects of Enabling Third-person Perspectives for Bystanders on VR-based Interactive Media**, CHI Workshop, 2019.
- 2018 **Student Leader**, ACI '18: Proceedings of the Fifth International Conference on Animal-Computer Interaction.

Volunteer and Community Services

- 2024 **Volunteer**, UbiComp '24 and ISWC 2024: ACM International Joint Conference on Pervasive and Ubiquitous Computing.
- 2023 **Volunteer**, The Tenth International Conference on Animal-Computer Interaction.
- 2021 **Volunteer**, ACM International Symposium on Wearable Computers.
- 2018-2019 **Health And Wellbeing Liaison**, The Campus Kitchens Project, Social Services, Georgia Institute of Technology.
- 2015-2018 **Shift Captain**, The Campus Kitchens Project, Social Services, Georgia Institute of Technology.

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